and T2, transformer X1, bridge rectifier BR1 and voltage regulator 7808 (IC5). The output of IC1 is fed to relay RL1 through the inverter built around transistors T1 and T2. If battery voltage goes above 5.5V, the circuit

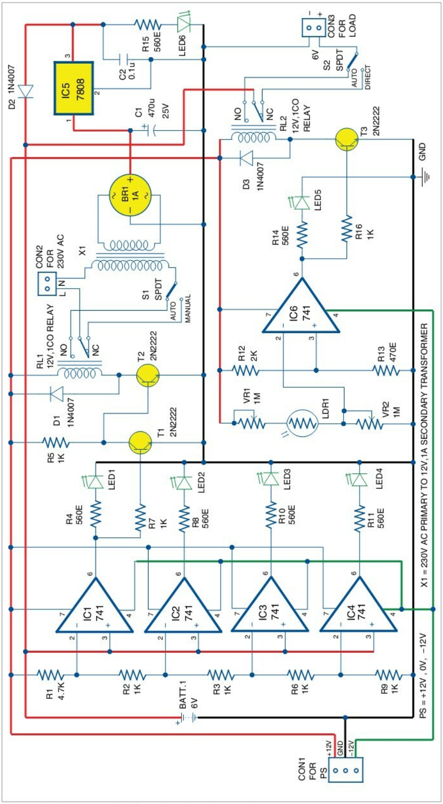

Fig. 2: Circuit for automatic light and overcharging control

PARTS LIST

Semiconductors:	
IC1-IC4, IC6	- 741 operation amplifier
IC5	- 7808, 8V voltage regulator
T1-T3	- 2N2222 npn transistor
LED1-LED6	- 5mm LED
D1-D3	- 1N4007 rectifier diode
BR1	- 1A, bridge rectifier
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 4.7-kilo-ohm
R2, R3, R5-R7, R9, R16	- 1-kilo-ohm
R4, R8, R10, R11, R14, R15	- 560-ohm
R12	- 2-kilo-ohm
R13	- 470-ohm
VR1, VR2	- 1-mega-ohm potmeter
LDR1	- LDR
Capacitors:	
C1	- 470µF, 25V electrolytic
C2	- 0.1µF ceramic disk
Miscellaneous:	
BATT.1	- 6V rechargeable battery
CON1	- 3-pin connector
CON2, CON3	- 2-pin terminal connector
S1, S2	- SPDT switch
RL1, RL2	- 12V single-changeover relay
X1	- 230V AC primary to 12V, 1A secondary transformer
	- ±12V power supply (PS)
	- 6V rechargeable battery with 2-pin terminal connector

de-energises relay RL1 to prevent overcharging. If voltage level is less than 5.5V, it energises RL1 to start battery charging.

Switch S1 is used to charge the battery in either manual or auto mode. In manual mode, it provides direct path to charge the battery. But this mode has a drawback: When you forget to switch off S1, the battery will get overcharged. In auto mode, the battery is charged on-load or off-load with overcharging protection. Table I shows LED indication and relay status for different battery voltage levels.

Diode D2 provides unidirectional signal flow for battery charging from 8V DC supply.

Automatic light control. This unit is built around IC6, light-dependent resistor LDR1, transistor T3 and relay RL2. LDR1 provides control voltage to comparator IC6. You can control the sensitivity of LDR1 using potmeters VR1 and VR2. As per the setting of light intensity in LDR1, relay RL2 energises through T3 to switch on the load.

Switch S2 is used to activate the load in either direct or auto mode.